Advanced Algorithms 2026 2026

Sample Solutions GHW 01

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1 Placing fire trucks (15 Points)

For a collection of team-truck pairs $G \subseteq [m]$, we say that a pair i is feasible for G if both the team and the truck paired by i are not yet assigned by G. The greedy procedure is then as follows. Initially, let $G = \emptyset$. Repeatedly add a feasible team-truck pair i to G that maximizes the marginal gain $f(G \cup \{i\}) - f(G)$. Terminate if there are no feasible placements.

We analyse the approximation guarantee of the obtained solution G. Let $O \subseteq [m]$ be an optimal collection of pairs, and denote by G_t the collection constructed by the greedy algorithm after step t. Every village covered by O is either already covered by G, or is covered by a pair $i \in O \setminus G$. This means that

$$\text{OPT} = f(O) \leq f(G) + \sum_{i \in O \setminus G} (f(G \cup \{i\}) - f(G)).$$

Hence, fix a pair $i \in O \setminus G$ not selected by G. Let $j \in G$ be the first chosen placement in G conflicting with i, i.e. i and j share a team or a truck. Note that such a j exists because G is maximal. Let t be the iteration in which the greedy algorithm chose j. Since coverage has diminishing marginal gain, we have

$$\begin{aligned} f(G \cup \{i\}) - f(G) &\leq f(G_{t-1} \cup \{i\}) - f(G_{t-1}) \\ &\leq f(G_{t-1} \cup \{j\}) - f(G_{t-1}) && \text{(since } i \text{ is a feasible pair for } G_{t-1}\text{)} \\ &= f(G_t) - f(G_{t-1}). \end{aligned}$$

Observe that each pair $j \in G$ conflicts with at most two pairs $i \in O \setminus G$: one using the same team and one using the same truck. So in total we obtain

$$\begin{aligned} \text{OPT} &\leq f(G) + \sum_{i \in O \setminus G} (f(G \cup \{i\}) - f(G)) \\ &\leq f(G) + 2 \sum_{t} (f(G_t) - f(G_{t-1})) \\ &= 3f(G). \end{aligned}$$

In other words, the greedy solution G covers at least $\text{OPT}/3$ people.

2 Water Quality Testing (25 Points)

1. Consider $n = 1$ laboratory with m devices that each test the same $t = 1$ contaminant. Say the delivery fee is $d_1 = m$ and each device costs $c_{1j} = 1$. Then the cheapest integral solution orders one device and costs $m + 1$. On the other hand, consider the fractional solution that sets $y_1 = x_{1j} = 1/m$ for every $j \in [m]$. This is a feasible solution to the LP and has cost $d_1 \cdot 1/m + \sum_{j \in [m]} c_{1j} \cdot 1/m = 2$. Thus, the integrality gap is at least $(m + 1)/2$, which is unbounded.

2. Consider an integral solution (x, y) to the original LP. For each contaminant $k \in [t]$, choose one laboratory i from which a selected device tests the contaminant k and set $z_{ik} = 1$. For all other laboratories i', set $z_{i'k} = 0$. Then all added constraints hold. In other words, every feasible solution extends to a feasible solution of the strengthened LP with the same cost, so the LP is a valid relaxation.

3. Let (x, y, z) be an optimal solution to the strengthened LP. Independently activate each laboratory i with probability y_i. If it is activated, independently select the device j with probability x_{ij}/y_i. Order the selected devices.

   We analyse the expected cost. The device $j \in [m]$ from laboratory $i \in [n]$ is selected with probability $y_i \cdot x_{ij}/y_i = x_{ij}$. The delivery fee of laboratory i is paid only if at least one device is selected, so with probability at most y_i. In total, the expected cost is thus

$$\mathbb{E}[\text{cost}] \leq \sum_i d_i y_i + \sum_{i,j} c_{ij} x_{ij} = \text{OPT}\,.$$

   Fix a contaminant $k \in [t]$. We analyse the probability that it is tested. Denote by q_{ik} the probability that the laboratory i supplies a device testing the contaminant k. For $y_i > 0$, we have

$$\begin{aligned} q_{ik} &= y_i \left(1 - \prod_{j:k\in S_{ij}} (1 - x_{ij}/y_i)\right) \\ &\geq y_i \left(1 - e^{-a_{ik}/y_i}\right) && (\text{for } a_{ik} \coloneqq \textstyle\sum_{j:k\in S_{ij}} x_{ij}) \\ &\geq \frac{y_i a_{ik}}{y_i + a_{ik}} \\ &\geq \frac{1}{2}\min\{y_i, a_{ik}\} \\ &\geq \frac{1}{2} z_{ik}. \end{aligned}$$

   Here we use $1 - x \leq e^{-x}$ and $1 - e^{-x} \geq \frac{x}{1+x}$ for $x \geq 0$. For $y_i = 0$, we have $z_{ik} = 0$, so $q_{ik} \geq \frac{1}{2} z_{ik}$ also holds. Since the laboratories are independently activates, we obtain

$$\Pr[k \text{ is not tested}] = \prod_i (1 - q_{ik}) \leq \prod_i e^{-q_{ik}} \leq e^{-\frac{1}{2}\Sigma_i z_{ik}} \leq e^{-1/2}$$

   where we use the constraint $1 \leq \Sigma_i z_{ik}$ for all $k \in [t]$. Hence, for each contaminant, the probability that it is tested is at least $1 - e^{-1/2}$.

4. In Question 3 we designed a procedure that selects devices such that for each contaminant, the probability that it is tested is constant. We now adapt the algorithm so that the selected devices test all contaminants with constant probability, i.e. the selection is a feasible solution. Later, we boost the success probability to $1 - \delta$.

   Let $A_1, \ldots, A_r$ denote the selected devices when running the randomized procedure from Question 3 independently $r = \lceil 2\ln(4t) \rceil$ times. Denote by B the union of the selected devices. Let $k \in [t]$ be a contaminant. From Question 3, for each $\ell \in [r]$ we know that

$$\Pr[k \text{ is not tested by } A_\ell] \leq e^{-1/2}.$$

   By independence, we infer that

$$\Pr[k \text{ is not tested by } B] \leq e^{-r/2} \leq \frac{1}{4t}.$$

By union bound, the probability that there exists a contaminant not tested by B is

$$\Pr[B \text{ is not feasible}] \leq \sum_{k=1}^{t} \Pr[k \text{ is not tested by } B] \leq \frac{1}{4}.$$

Since $\mathbb{E}[\text{cost}(A_\ell)] \leq \text{OPT}$ by Question 3, we have $\mathbb{E}[\text{cost}(B)] \leq r\,\text{OPT}$. By Markov's inequality,

$$\Pr[\text{cost}(B) > 4r\,\text{OPT}] \leq 1/4.$$

We say that B is successful if it is feasible and its cost is at most $4r\,\text{OPT}$. This happens with probability at least

$$\Pr[B \text{ is successful}] \geq 1 - \Pr[B \text{ is not feasible}] - \Pr[\text{cost}(B) > 4r\,\text{OPT}] \geq 1/2.$$

To boost the success probability, repeat the experiment B independently $q = \lceil \log(1/\delta) \rceil$ times and return the cheapest feasible result. Since each experiment is not successful with probability at most $1/2$, the probability that all experiments are not successful is at most $(1/2)^q \leq \delta$. In other words, with probability at least $1 - \delta$, the procedure selects devices that test all contaminants and that cost at most $4r\,\text{OPT} = \mathcal{O}(\log t)\,\text{OPT}$.

3 Scheduling jobs with quadratic load cost (30 points)

This exercise was inspired byan exercise in the tutorial by Petra Schuurman and Gerhard Woeginger (see here).

1. Let $s_1, \ldots, s_C$ be the distinct processing time. We denote by n_i the number of jobs of size s_i. For an integer vector $v \in [n_1] \times \cdots \times [n_C]$, let $D(t, v)$ be the minimum cost of placing v_i jobs of size s_i on t machines. Set $D(0, 0) = 0$ and $D(0, v) = +\infty$ for $v \neq 0$, and use

$$D(t, v) = \min_{0 \leq u \leq v} \left\{ D(t-1, v-u) + \left(\sum_{i=1}^{C} u_i s_i \right)^2 \right\}$$

where vector inequalities are component-wise. The vector u specifies exactly the jobs on the last machine, so the recurrence considers every possible assignment. The optimal cost is given in $D(m, (n_1, \ldots, n_C)) = \text{OPT}$. There are at most $\mathcal{O}(mn^C)$ states and in each transition we consider at most $\mathcal{O}(n^C)$ candidate vectors u. In total, the algorithm runs in time $\mathcal{O}(mn^{2C})$, which is polynomial for fixed C.

2. Let $P \coloneqq \sum_j p_j$ be the total processing time and $A \coloneqq P/m$. First, we argue that in an optimal schedule, every job with processing time larger than A occupies its own machine. Indeed, suppose it shares a machine of load y with another job of processing time $s > 0$. Then there is a machine of load $x < A$. Moving the job of size s to that machine changes the cost by

$$(x+s)^2 + (y-s)^2 - x^2 - y^2 = 2s(x - y + s) < 0,$$

since $y - s > A \geq x$. This contradicts optimality. Since we know how to optimally schedule jobs with size greater than A, we can assume without loss of generality that every job has size at most A.

Solve a rounded instance. We now describe a procedure that rounds the processing times to obtain only a constant number of distinct processing times. Let $h = \delta A$ and $g = \delta h$, for a parameter $\delta \in (0, 1)$ depending on ε which we will tune at the end. A job is *small* if its size is at most h, and *big* otherwise. Let S be the total processing time of small jobs. Replace them by $q = \lfloor S/h \rfloor$ artificial jobs of size h. Note that this doesn't increase the number of jobs. For every big job p_j, round its size up to $\widehat{p}_j = g\lceil p_j/g \rceil$. Each big job has $p_j \leq A$, so its rounded size $\hat{p}_j$ belongs to $\{g, 2g, \ldots, \lceil A/g \rceil g\}$. There are at most $\lceil A/g \rceil = \lceil \delta^{-2} \rceil$ such sizes. Since all artificial small jobs have the same size, the constructed instance has $C \leq \mathcal{O}(\delta^{-2})$ distinct sizes.

We can thus solve the constructed instance using the above DP and obtain a schedule where the machine i has load $L_i^{\#}$. Additionally, we notice that the rounded instance has at most n jobs, and the total processing time of the rounded jobs is at most $\sum_i L_i^{\#} \leq (1+\delta)P$.

Restore a schedule of the original jobs. We modify the obtain schedule to accommodate for the original jobs. Every big job is scheduled according to the obtained schedule: since $p_j \leq \hat{p}_j$, this doesn't increase the machine loads. For small jobs, we proceed as follows. If machine i received q_i artificial small jobs, remove them and reserve $q_i h + 2h$ units of load for original small jobs. Use First Fit to place the small jobs in the reserved units, i.e. place each small job in the first reservation with enough space. We argue that we reserved enough space for every small job *(repeat the First-Fit correctness argument)*. Indeed, if a job did not fit, every reservation would have less than h space left. The small-job volume already placed would then exceed

$$\sum_{i=1}^{m}(q_i h + h) = qh + mh \geq qh + h > S,$$

which is impossible. Thus, all jobs fit, and each final load is at most $L_i^{\#} + 2h$.

Consequently, the resulting schedule has a cost of

$$ALG \leq \sum_{i=1}^{m}(L_i^{\#} + 2h)^2 = \sum_{i=1}^{m}(L_i^{\#})^2 + 4h\sum_{i=1}^{m} L_i^{\#} + 4mh^2. \tag{1}$$

Note that the first term $Q = \sum_i (L_i^{\#})^2$ is the optimal cost of the rounded instance given by the DP. We now bound Q. Consider an optimal schedule on the original instance and L_i^* be the load of machine i. Denote S_i the total size of small jobs scheduled on machine i. We construct a schedule of the rounded instance with loads L_i' as follows. On each machine, replace the volume S_i of small jobs by at most $\lceil S_i/h \rceil$ artificial jobs of size h. There is room for all q artificial jobs, because $\sum_i \lceil S_i/h \rceil \geq S/h \geq q$. Thus, the artificial small-job load on machine i is at most $h\lceil S_i/h \rceil \leq S_i + h$.

For the big jobs, notice that their total volume is $L_i^* - S_i$. We replace each big jobs of size p_j by its rounded size $\hat{p}_j$. Note that since $p_j > h$ and $g = \delta h$, we have $\widehat{p}_j \leq p_j + g \leq (1+\delta)p_j$. So the volume taken by the rounded big jobs is at most $(1 + \delta) \cdot (L_i^* - S_i)$. In total, the constructed schedule has load at most

$$L_i' \leq (1 + \delta)(L_i^* - S_i) + S_i + h \leq (1 + \delta)L_i^* + h$$

on machine i. Since this schedule is feasible on the rounded instance, we get

$$\begin{aligned}
Q &\le \sum_{i=1}^{m} \big((1+\delta)L_i^* + h\big)^2 \\
&= (1+\delta)^2 \sum_{i=1}^{m} (L_i^*)^2 + 2(1+\delta)h \sum_{i=1}^{m} L_i^* + \sum_{i=1}^{m} h^2 \\
&= (1+\delta)^2 \,\mathrm{OPT} + 2(1+\delta)hP + mh^2 \\
&\le \big((1+\delta)^2 + 2\delta(1+\delta) + \delta^2\big)\,\mathrm{OPT} = (1+2\delta)^2\,\mathrm{OPT},
\end{aligned}$$

Here we use the observation that by Cauchy-Schwarz, $P^2 = (\sum_i L_i^*)^2 \le m \sum_i (L_i^*)^2 = m\,\mathrm{OPT}$ and thus $mA^2 \le \mathrm{OPT}$, $hP = (\delta A)(mA) = \delta m A^2 \le \delta\,\mathrm{OPT}$ and $mh^2 = m(\delta A)^2 = \delta^2 m A^2 \le \delta^2\,\mathrm{OPT}$.

We can now bound the cost in (1). Remember that the total processing time of the rounded instance increased by a factor at most $(1+\delta)$, thus

$$\begin{aligned}
ALG &\le Q + 4h \sum_i L_i^{\#} + 4mh^2 \\
&\le (1+2\delta)^2\,\mathrm{OPT} + 4\delta(1+\delta)\,\mathrm{OPT} + 4\delta^2\,\mathrm{OPT} \\
&= (1+8\delta+12\delta^2)\,\mathrm{OPT} \le (1+4\delta)^2\,\mathrm{OPT}.
\end{aligned}$$

Choosing δ and running time. For $\varepsilon \in (0,1)$, choose $\delta = \varepsilon/12$. Then

$$(1+4\delta)^2 = 1 + \frac{2}{3}\varepsilon + \frac{1}{9}\varepsilon^2 \le 1 + \varepsilon.$$

The rounded instance has $C = O(\varepsilon^{-2})$ distinct processing times. The DP-call takes time $O(mn^{2C}) = \mathcal{O}(mn^{c/\varepsilon^2})$ for some constant $c > 0$; First Fit takes time $O(nm)$.

4 Currency exchange (30 points)

This exercise was inspired by an exercise in the Final Exam, of the Advanced Algorithms class taught in 2018-2019 by Ola Svensson at EPFL.

1. Let $M = 2^k$ and $h = \sqrt{M}$. Exchange at the first rate at least h, or on the last day if no earlier rate reaches h. If some rate reaches h, the revenue is at least h, while $\mathrm{OPT} \le M$, giving ratio at most $M/h = h$. Otherwise, $\mathrm{OPT} < h$ and the revenue is at least one, again giving ratio at most h.

   We design an example on two days (i.e. $T = 2$). Set $x_1 = h$. If the deterministic algorithm exchanges on day 1, set $x_2 = h^2$. If the deterministic algorithm exchanges on day 2, set $x_2 = 1$. Both give ratio h.

2. Choose J uniformly from $\{0, \dots, k-1\}$, independently of the rates. Exchange at the first rate at least 2^J, or on the last day otherwise. For any fixed sequence with maximum rate P, there is an index $j^* \in \{0, \dots, k-1\}$ with $P/2 \le 2^{j^*} \le P$. With probability $1/k$, we choose this index and receive at least $P/2$ Swiss francs. Consequently, the expected revenue is at least $Pr[J = j^*] \cdot P/2 = \mathrm{OPT}/(2k)$.

3. Set $T = k + 1$ and, for $1 \leq j \leq k$, consider the sequence of rates

$$\sigma_j = (2, 2^2, \ldots, 2^j, \underbrace{1, \ldots, 1}_{k+1-j \text{ entries}}).$$

All sequences have the same announced length, and $\mathrm{OPT}(\sigma_j) = 2^j$.

Fix a deterministic algorithm A. Let $i \in [T]$ be its exchange day when given the sequence of rates σ_k. We analyse the revenue of A when given the sequence σ_j for any $j \in [k]$. If $j \geq i$, then the first i rates in σ_k and σ_j are identical and thus the algorithm A makes the same decision to exchange on day i, i.e. it receives 2^i Swiss francs. If $j < i$, then the rates on the first j days in σ_k and σ_j are identical and the algorithm A makes the same decision to not exchange on day j (or earlier). This means that it exchanges on a later day and thus receives only 1 Swiss franc.

Summing the competitive ratio over every sequence σ_j gives

$$\sum_{j=1}^{k} \frac{A(\sigma_j)}{2^j} = \sum_{j=1}^{i-1} 2^{-j} + \sum_{j=i}^{k} 2^{i-j} \leq 1 + 2 = 3.$$

Intuitively, the algorithm doesn't perform well on too many sequences of the family $\{\sigma_j\}_{j \in [k]}$.

Now let R be any randomized algorithm. Fixing its entire random tape gives a deterministic algorithm, for which the above inequality holds. In particular, by linearity of expectation: $\sum_{j=1}^{k} \frac{\mathbb{E}[R(\sigma_j)]}{\mathrm{OPT}(\sigma_j)} \leq 3$. There exists a choice of $j \in [k]$ such that

$$\frac{\mathbb{E}[R(\sigma_j)]}{\mathrm{OPT}(\sigma_j)} \leq \frac{1}{k} \sum_{j=1}^{k} \frac{\mathbb{E}[R(\sigma_j)]}{\mathrm{OPT}(\sigma_j)} \leq \frac{3}{k},$$

thus proving a competitive ratio of at least $k/3$. Note that the choice of j depends only on the algorithm and its probability distribution, not on the outcome of any particular run, and is thus indeed an oblivious adversary.